Program related

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1. Commencement Phase – Core Mechanics Foundation

Focus: Driving physics architecture + steering system prototype

Deliverables:

1.1 Driving Physics Framework (F1-Centric)

- Custom vehicle controller system (not default physics reliance)
- Downforce simulation scaling with speed
- Grip coefficient system (track surface + speed based)
- High-speed stability logic implementation
- Tire traction curves (low, medium, high speed bands)
- Early weight transfer simulation (acceleration, braking, cornering)

Acceptance Criteria:

- Car remains stable at high speeds (no unwanted spinouts)
- Predictable turning response at varying speeds
- Physics debug panel for tuning variables

1.2 Predictive & Fluid Steering – Prototype Version

- Steering smoothing system
- Speed-based steering sensitivity scaling

- Initial steering assist (light autocorrect)
- Oversteer detection prototype
- Understeer compensation logic

Acceptance Criteria:

- No snapping or jerky steering
- Steering input feels continuous and progressive
- Basic autocorrect reduces extreme fishtailing

1.3 Camera Speed Perception – First Pass

- Third-person follow camera system
- Dynamic FOV scaling based on speed
- Camera lag tuning for weight sensation
- Initial shake prototype (optional toggle)

Acceptance Criteria:

- Strong speed sensation at high velocity
- No camera jitter
- Smooth follow behavior

1.4 Internal Physics Test Track

- Basic test circuit for tuning

- Straight-line speed test section
- High-speed corner testing zone
- Heavy braking zone

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2. Midpoint Review (~Month 3) – Refined Handling & AI Integration

Focus: Advanced steering refinement + braking mechanics + AI first pass

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Deliverables:

2.1 Advanced Steering & Autocorrect System

- Dynamic steering correction based on slip angle
- Traction loss detection
- Adjustable driver assist levels (Low / Medium / High)
- Reduced “arcade feel” steering
- Countersteer logic tuning

Acceptance Criteria:

- Controlled recovery from oversteer
- Predictive steering feel at high speed
- Steering assist does not override skilled input

2.2 Late Braking & Weight Transfer System

- Braking force curve tuning
- Front weight shift simulation under braking
- Rear instability simulation under heavy brake
- Trail braking support
- Lockup threshold simulation

Acceptance Criteria:

- Player can brake late without instant spin
- Car nose dips under braking
- Brake modulation matters

2.3 AI Behaviour – Version 1

- Racing line system
- Corner speed adaptation
- Collision avoidance logic
- Basic overtaking logic
- Difficulty scaling presets

Acceptance Criteria:

- AI follows racing line consistently
- No aggressive ramming
- Competitive but fair overtakes

2.4 Handling Tuning Documentation

- Physics variable documentation
- Steering sensitivity ranges
- Brake force calibration sheet
- Downforce scaling chart

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3. Final Development Phase (~Month 6) – Polished Racing Feel

Focus: Competitive polish + stability + realism tuning

Deliverables:

3.1 Final Steering System (Production Version)

- Fully tuned predictive steering
- Finalized autocorrect strength curve
- Slip recovery smoothing
- Advanced traction management refinement
- Steering input deadzone calibration

Acceptance Criteria:

- Feels “confident” at 200+ speed
- No unnatural snap corrections
- Balanced skill ceiling (assist doesn’t dominate)

3.2 Final Downforce & Grip Calibration

- Speed-to-downforce curve finalized
- High-speed corner stability confirmed
- Realistic loss of grip at low speeds
- Surface grip modifiers (if applicable)

3.3 AI Behaviour – Final Version

- Adaptive racing logic
- Clean defensive driving behavior
- Slipstream logic (if implemented)
- Consistent lap time targeting

Acceptance Criteria:

- AI behaves like confident F1 drivers
- No erratic braking
- Competitive across difficulty tiers

3.4 Camera & Speed Perception – Final Pass

- Polished dynamic FOV
- Speed blur (if implemented)
- High-speed stability lock

- Cinematic replay camera prototype (optional stretch goal)

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4. Completion Review & Submission Readiness (Month 7–9)

Focus: Optimization + QA + Delivery Package

Deliverables:

4.1 Optimization

- Physics performance profiling
- Stable framerate under race conditions
- AI CPU load optimization
- Network sync testing (if multiplayer)

4.2 Bug Fix & Stability Pass

- Steering edge-case fixes
- Brake system edge-case fixes
- AI pathing corrections
- Camera clipping fixes

4.3 Final Build Delivery

- Production-ready game build

- Physics tuning file
- AI configuration file
- Handling documentation
- Internal testing report

4.4 Client Review Session

- Live demo session
- Adjustment feedback window
- Final revision pass (agreed scope only)